

## 1. General Description

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The EM74HC259; EM74HCT259 is an 8-bit addressable latch. The device features four modes of operation. In the addressable latch mode, data on the D input is written into the latch addressed by the inputs A0 to A3. The addressed latch will follow the data input, non-addressed latches will retain their previous states. In memory mode, all latches retain their previous states and are unaffected by the data or address inputs. In the 3-to-8 decoding or demultiplexing mode, the addressed output follows the D input and all other outputs are LOW. In the reset mode, all outputs are forced LOW and unaffected by the data or address inputs. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of  $V_{CC}$ .

## 2. Features and Benefits

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- Wide supply voltage range from 2.0 V to 6.0 V
- Latch-up performance exceeds 250 mA
- Complies with JEDEC standards:
  - JESD8C (2.7 V to 3.6 V)
  - JESD7A (2.0 V to 6.0 V)
- Combined demultiplexer and 8-bit latch
- Serial-to-parallel capability
- Output from each storage bit available
- Random (addressable) data entry
- Easily expandable
- Common reset input
- Useful as a 3-to-8 active HIGH decoder
- Input levels:
  - For EM74HC259: CMOS level
  - For EM74HCT259: TTL level

■ ESD protection:

- HBM: ANSI/ESDA/JEDEC JS-001 class 2 exceeds 3500 V
- CDM: ANSI/ESDA/JEDEC JS-002 class C3 exceeds 2000 V

■ Multiple package options

■ Specified from -40 °C to +85 °C and from -40 °C to +125 °C

## 3. Ordering Information

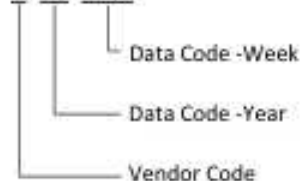
Table 1. Ordering information

| Type number  | Topside marking | Package   |                                                                           |          |
|--------------|-----------------|-----------|---------------------------------------------------------------------------|----------|
|              |                 | Name      | Description                                                               | Quantity |
| EM74HC259D   | HC259<br>XYWW   | SOP-16L   | plastic small outline package; 16 leads;<br>body width 3.9 mm             | 3000     |
| EM74HCT259D  | HCT259<br>XYWW  |           |                                                                           |          |
| EM74HC259PW  | HC259<br>XYWW   | TSSOP-16L | plastic thin shrink small outline package;<br>16 leads; body width 4.4 mm | 3000     |
| EM74HCT259PW | HCT259<br>XYWW  |           |                                                                           |          |

### MARKING INFORMATION

NOTE: XYWW = Vendor Code and Data Code.

X YY WW



## 4. Function Diagram

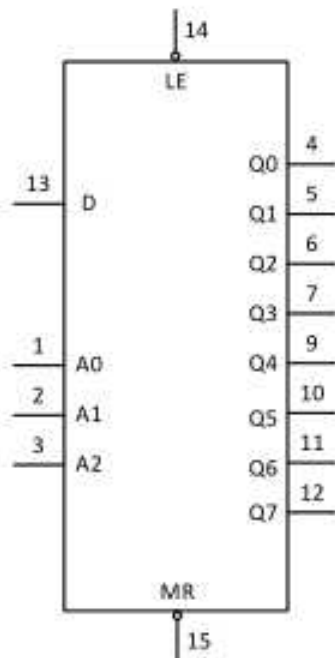


Fig. 1. Logic symbol

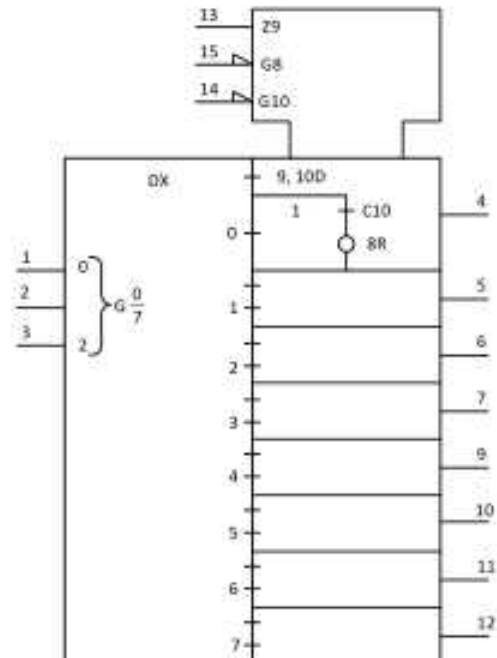


Fig. 2. IEC logic symbol

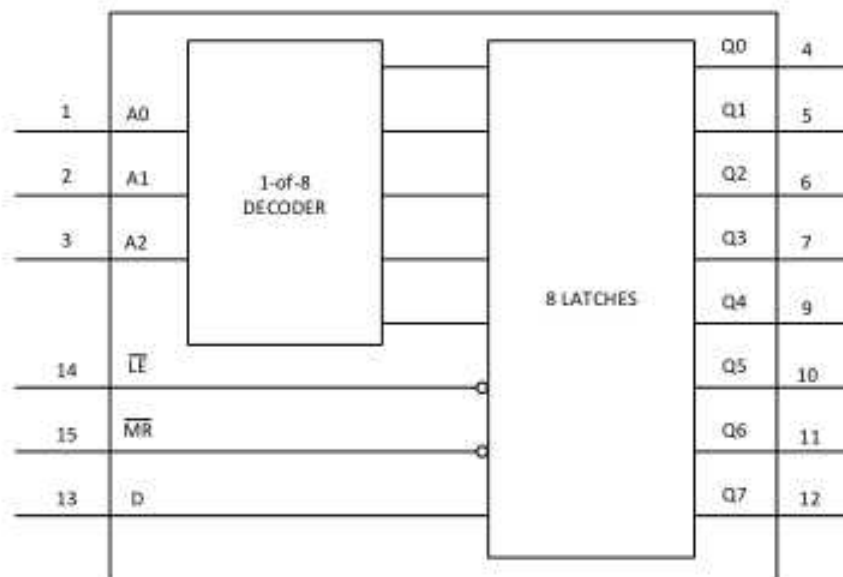


Fig. 3. Functional diagram

## 5. Pinning Information

### 5.1. Pinning

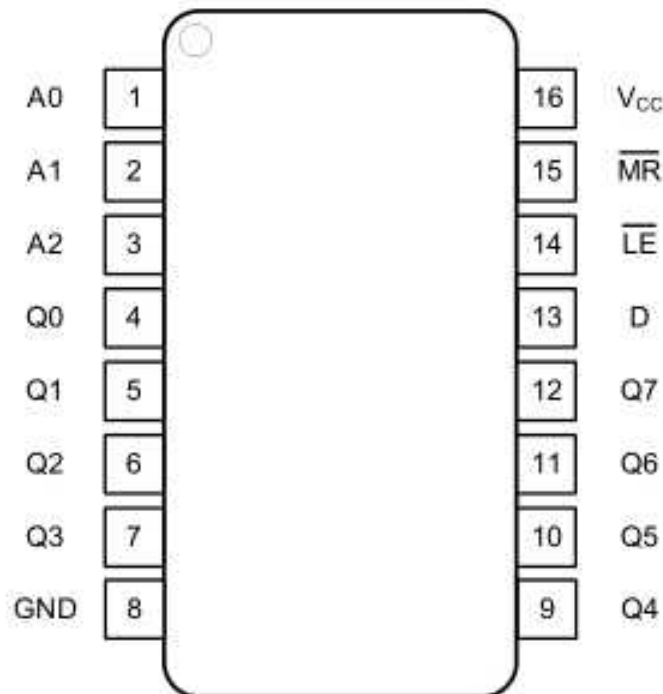


Fig. 4. Top view pin configuration SOP and TSSOP

### 5.2. Pin description

Table 2. Pin description

| Symbol                         | Pin                       | Description                          |
|--------------------------------|---------------------------|--------------------------------------|
| A0, A1, A2                     | 1, 2, 3                   | Address input                        |
| Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7 | 4, 5, 6, 7, 9, 10, 11, 12 | Latch output                         |
| GND                            | 8                         | Ground (0 V)                         |
| D                              | 13                        | Data input                           |
| $\overline{LE}$                | 14                        | Latch enable input (active LOW)      |
| $\overline{MR}$                | 15                        | Conditional reset input (active LOW) |
| V <sub>CC</sub>                | 16                        | Supply voltage                       |

## 6. Functional Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level; X = don't care;

d = HIGH or LOW data one set-up time prior to the LOW-to-HIGH LE transition;

q = lower case letter indicates the state of the referenced input one set-up time prior to the LOW-to-HIGH transition.

| Operating mode                                             | Input                  |                        |   |    |    |    | Output         |                |                |                |                |                |                |                |
|------------------------------------------------------------|------------------------|------------------------|---|----|----|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|                                                            | $\overline{\text{MR}}$ | $\overline{\text{LE}}$ | D | A0 | A1 | A2 | Q0             | Q1             | Q2             | Q3             | Q4             | Q5             | Q6             | Q7             |
| Reset (clear)                                              | L                      | H                      | X | X  | X  | X  | L              | L              | L              | L              | L              | L              | L              | L              |
| Demultiplexer (active HIGH 8-channel) decoder (when D = H) | L                      | L                      | d | L  | L  | L  | Q=d            | L              | L              | L              | L              | L              | L              | L              |
|                                                            | L                      | L                      | d | H  | L  | L  | L              | Q=d            | L              | L              | L              | L              | L              | L              |
|                                                            | L                      | L                      | d | L  | H  | L  | L              | L              | Q=d            | L              | L              | L              | L              | L              |
|                                                            | L                      | L                      | d | H  | H  | L  | L              | L              | L              | Q=d            | L              | L              | L              | L              |
|                                                            | L                      | L                      | d | L  | L  | H  | L              | L              | L              | L              | Q=d            | L              | L              | L              |
|                                                            | L                      | L                      | d | H  | L  | H  | L              | L              | L              | L              | L              | Q=d            | L              | L              |
|                                                            | L                      | L                      | d | L  | H  | H  | L              | L              | L              | L              | L              | L              | Q=d            | L              |
|                                                            | L                      | L                      | d | H  | H  | H  | L              | L              | L              | L              | L              | L              | L              | Q=d            |
| Memory (no action)                                         | H                      | H                      | X | X  | X  | X  | q <sub>0</sub> | q <sub>1</sub> | q <sub>2</sub> | q <sub>3</sub> | q <sub>4</sub> | q <sub>5</sub> | q <sub>6</sub> | q <sub>7</sub> |
| Addressable latch                                          | H                      | L                      | d | L  | L  | L  | Q=d            | q <sub>1</sub> | q <sub>2</sub> | q <sub>3</sub> | q <sub>4</sub> | q <sub>5</sub> | q <sub>6</sub> | q <sub>7</sub> |
|                                                            | H                      | L                      | d | H  | L  | L  | q <sub>0</sub> | Q=d            | q <sub>2</sub> | q <sub>3</sub> | q <sub>4</sub> | q <sub>5</sub> | q <sub>6</sub> | q <sub>7</sub> |
|                                                            | H                      | L                      | d | L  | H  | L  | q <sub>0</sub> | q <sub>1</sub> | Q=d            | q <sub>3</sub> | q <sub>4</sub> | q <sub>5</sub> | q <sub>6</sub> | q <sub>7</sub> |
|                                                            | H                      | L                      | d | H  | H  | L  | q <sub>0</sub> | q <sub>1</sub> | q <sub>2</sub> | Q=d            | q <sub>4</sub> | q <sub>5</sub> | q <sub>6</sub> | q <sub>7</sub> |
|                                                            | H                      | L                      | d | L  | L  | H  | q <sub>0</sub> | q <sub>1</sub> | q <sub>2</sub> | q <sub>3</sub> | Q=d            | q <sub>5</sub> | q <sub>6</sub> | q <sub>7</sub> |
|                                                            | H                      | L                      | d | H  | L  | H  | q <sub>0</sub> | q <sub>1</sub> | q <sub>2</sub> | q <sub>3</sub> | q <sub>4</sub> | Q=d            | q <sub>6</sub> | q <sub>7</sub> |
|                                                            | H                      | L                      | d | L  | H  | H  | q <sub>0</sub> | q <sub>1</sub> | q <sub>2</sub> | q <sub>3</sub> | q <sub>4</sub> | q <sub>5</sub> | Q=d            | q <sub>7</sub> |
|                                                            | H                      | L                      | d | H  | H  | H  | q <sub>0</sub> | q <sub>1</sub> | q <sub>2</sub> | q <sub>3</sub> | q <sub>4</sub> | q <sub>5</sub> | q <sub>6</sub> | Q=d            |

**Table 4. Operating mode select table**

H = HIGH voltage level; L = LOW voltage level.

| $\overline{\text{LE}}$ | $\overline{\text{MR}}$ | Mode                   |
|------------------------|------------------------|------------------------|
| L                      | H                      | Addressable latch mode |
| H                      | H                      | Memory mode            |
| L                      | L                      | Demultiplexer mode     |
| H                      | L                      | Reset mode             |

## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                                                                | Min  | Max      | Unit               |
|-----------|-------------------------|---------------------------------------------------------------------------|------|----------|--------------------|
| $V_{CC}$  | supply voltage          |                                                                           | -0.5 | 7        | V                  |
| $I_{IK}$  | input clamping current  | $V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$                    |      | $\pm 20$ | mA                 |
| $I_{OK}$  | output clamping current | $V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$                    |      | $\pm 20$ | mA                 |
| $I_O$     | output current          | $V_O = -0.5\text{ V}$ to $(V_{CC} + 0.5\text{ V})$                        |      | $\pm 25$ | mA                 |
| $I_{CC}$  | supply current          |                                                                           |      | 70       | mA                 |
| $I_{GND}$ | ground current          |                                                                           | -70  |          | mA                 |
| $P_{tot}$ | total power dissipation | $T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$ |      | 500      | mW                 |
| $T_{stg}$ | storage temperature     |                                                                           | -65  | 150      | $^{\circ}\text{C}$ |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol              | Parameter                           | Conditions              | EM74HC259 |      |          | EM74HCT259 |      |          | Unit               |
|---------------------|-------------------------------------|-------------------------|-----------|------|----------|------------|------|----------|--------------------|
|                     |                                     |                         | Min       | Typ  | Max      | Min        | Typ  | Max      |                    |
| $V_{CC}$            | supply voltage                      |                         | 2.0       | 5.0  | 6.0      | 4.5        | 5.0  | 5.5      | V                  |
| $V_I$               | input voltage                       |                         | 0         |      | $V_{CC}$ | 0          |      | $V_{CC}$ | V                  |
| $V_O$               | output voltage                      |                         | 0         |      | $V_{CC}$ | 0          |      | $V_{CC}$ | V                  |
| $T_{amb}$           | ambient temperature                 |                         | -40       | 25   | 125      | -40        | 25   | 125      | $^{\circ}\text{C}$ |
| $\Delta t/\Delta V$ | Input transition rise and fall rate | $V_{CC} = 2.0\text{ V}$ |           |      | 625      |            |      |          | ns/V               |
|                     |                                     | $V_{CC} = 4.5\text{ V}$ |           | 1.67 | 139      |            | 1.67 | 139      | ns/V               |
|                     |                                     | $V_{CC} = 6.0\text{ V}$ |           |      | 83       |            |      |          | ns/V               |

## 9.Static Characteristics

**Table 6. Static characteristics**

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

| Symbol          | Parameter                 | Conditions                                                                                  | -40 °C to +85 °C |     |      | -40 °C to +125 °C |      | Unit |
|-----------------|---------------------------|---------------------------------------------------------------------------------------------|------------------|-----|------|-------------------|------|------|
|                 |                           |                                                                                             | Min              | Typ | Max  | Min               | Max  |      |
| EM74HC259       |                           |                                                                                             |                  |     |      |                   |      |      |
| V <sub>IH</sub> | HIGH-level input voltage  | V <sub>CC</sub> = 2.0 V                                                                     | 1.5              |     |      | 1.5               |      | V    |
|                 |                           | V <sub>CC</sub> = 4.5 V                                                                     | 3.15             |     |      | 3.15              |      | V    |
|                 |                           | V <sub>CC</sub> = 6.0 V                                                                     | 4.2              |     |      | 4.2               |      | V    |
| V <sub>IL</sub> | LOW-level input voltage   | V <sub>CC</sub> = 2.0 V                                                                     |                  |     | 0.5  |                   | 0.5  | V    |
|                 |                           | V <sub>CC</sub> = 4.5 V                                                                     |                  |     | 1.35 |                   | 1.35 | V    |
|                 |                           | V <sub>CC</sub> = 6.0 V                                                                     |                  |     | 1.8  |                   | 1.8  | V    |
| V <sub>OH</sub> | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                         |                  |     |      |                   |      |      |
|                 |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 2.0 V                                            | 1.9              |     |      | 1.9               |      | V    |
|                 |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 4.5 V                                            | 4.4              |     |      | 4.4               |      | V    |
|                 |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 6.0 V                                            | 5.9              |     |      | 5.9               |      | V    |
|                 |                           | I <sub>O</sub> = -4.0 mA; V <sub>CC</sub> = 4.5 V                                           | 3.84             |     |      | 3.7               |      | V    |
|                 |                           | I <sub>O</sub> = -5.2 mA; V <sub>CC</sub> = 6.0 V                                           | 5.34             |     |      | 5.2               |      | V    |
| V <sub>OL</sub> | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                         |                  |     |      |                   |      |      |
|                 |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 2.0 V                                             |                  |     | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 4.5 V                                             |                  |     | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 6.0 V                                             |                  |     | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 4.0 mA; V <sub>CC</sub> = 4.5 V                                            |                  |     | 0.33 |                   | 0.4  | V    |
|                 |                           | I <sub>O</sub> = 5.2 mA; V <sub>CC</sub> = 6.0 V                                            |                  |     | 0.33 |                   | 0.4  | V    |
| I <sub>i</sub>  | input leakage current     | V <sub>i</sub> = V <sub>CC</sub> or GND ;<br>V <sub>CC</sub> = 6.0 V                        |                  |     | ±1   |                   | ±1   | μA   |
| I <sub>CC</sub> | supply current            | V <sub>i</sub> = V <sub>CC</sub> or GND ; I <sub>O</sub> = 0 A ;<br>V <sub>CC</sub> = 6.0 V |                  |     | 20   |                   | 40   | μA   |
| C <sub>i</sub>  | input capacitance         |                                                                                             |                  | 6.5 |      |                   |      | pF   |

# EM74HC259; EM74HCT259

8-bit addressable latch

| Symbol           | Parameter                 | Conditions                                                                                                                               | -40 °C to +85 °C |     |      | -40 °C to +125 °C |     | Unit |
|------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|------|-------------------|-----|------|
|                  |                           |                                                                                                                                          | Min              | Typ | Max  | Min               | Max |      |
| EM74HCT259       |                           |                                                                                                                                          |                  |     |      |                   |     |      |
| V <sub>IH</sub>  | HIGH-level input voltage  | V <sub>CC</sub> = 4.5 V to 5.5 V                                                                                                         | 2.0              |     |      | 2.0               |     | V    |
| V <sub>IL</sub>  | LOW-level input voltage   | V <sub>CC</sub> = 4.5 V to 5.5 V                                                                                                         |                  |     | 0.8  |                   | 0.8 | V    |
| V <sub>OH</sub>  | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ; V <sub>CC</sub> = 4.5 V                                                            |                  |     |      |                   |     |      |
|                  |                           | I <sub>O</sub> = -20 μA                                                                                                                  | 4.4              |     | 4.4  |                   | V   |      |
|                  |                           | I <sub>O</sub> = -4.0 mA                                                                                                                 | 3.84             |     | 3.7  |                   | V   |      |
| V <sub>OL</sub>  | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ; V <sub>CC</sub> = 4.5 V                                                            |                  |     |      |                   |     |      |
|                  |                           | I <sub>O</sub> = 20 μA                                                                                                                   |                  |     | 0.1  |                   | 0.1 | V    |
|                  |                           | I <sub>O</sub> = 4.0 mA                                                                                                                  |                  |     | 0.33 |                   | 0.4 | V    |
| I <sub>i</sub>   | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ; V <sub>CC</sub> = 5.5 V                                                                        |                  |     | ±1   |                   | ±1  | μA   |
| I <sub>CC</sub>  | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND ; I <sub>O</sub> = 0 A ; V <sub>CC</sub> = 5.5 V                                                 |                  |     | 20   |                   | 40  | μA   |
| ΔI <sub>CC</sub> | additional supply current | V <sub>I</sub> = V <sub>CC</sub> - 2.1 V; other inputs at V <sub>CC</sub> or GND; V <sub>CC</sub> = 4.5 V to 5.5 V; I <sub>O</sub> = 0 A |                  |     | 450  |                   | 490 | μA   |
| C <sub>i</sub>   | input capacitance         |                                                                                                                                          |                  | 6.5 |      |                   |     | pF   |



## 10. Dynamic Characteristics

**Table 7. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 11.

| Symbol           | Parameter                     | Conditions                                                | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|------------------|-------------------------------|-----------------------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|                  |                               |                                                           | Min              | Typ[1] | Max | Min               | Max |      |
| EM74HC259        |                               |                                                           |                  |        |     |                   |     |      |
| t <sub>pd</sub>  | propagation delay             | D to Q <sub>n</sub> ; see Fig. 5 [2]                      |                  |        |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                   |                  |        | 55  |                   | 60  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                   |                  |        | 17  |                   | 20  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                   |                  |        | 13  |                   | 15  | ns   |
|                  |                               | A <sub>n</sub> to Q <sub>n</sub> ; see Fig. 6 [2]         |                  |        |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                   |                  |        | 55  |                   | 60  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                   |                  |        | 17  |                   | 20  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                   |                  |        | 13  |                   | 15  | ns   |
|                  |                               | $\overline{\text{LE}}$ to Q <sub>n</sub> ; see Fig. 7 [2] |                  |        |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                   |                  |        | 55  |                   | 60  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                   |                  |        | 17  |                   | 20  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                   |                  |        | 13  |                   | 15  | ns   |
| t <sub>PHL</sub> | HIGH to LOW propagation delay | $\overline{\text{MR}}$ to Q <sub>n</sub> ; see Fig. 8     |                  |        |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                   |                  |        | 55  |                   | 60  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                   |                  |        | 17  |                   | 20  | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                   |                  |        | 13  |                   | 15  | ns   |
| t <sub>s</sub>   | transition time               | see Fig. 7 [3]                                            |                  |        |     |                   |     |      |
|                  |                               | V <sub>CC</sub> = 2.0 V                                   |                  |        | 8   |                   | 10  | ns   |
|                  |                               | V <sub>CC</sub> = 4.5 V                                   |                  |        | 7   |                   | 9   | ns   |
|                  |                               | V <sub>CC</sub> = 6.0 V                                   |                  |        | 7   |                   | 9   | ns   |

**EM74HC259; EM74HCT259**
**8-bit addressable latch**

| Symbol            | Parameter                     | Conditions                                             | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|-------------------|-------------------------------|--------------------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|                   |                               |                                                        | Min              | Typ[1] | Max | Min               | Max |      |
| $t_{pw}$          | pulse width                   | $\overline{LE}$ HIGH or LOW; see Fig. 7                |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 2.0\text{ V}$                                | 100              |        |     | 120               |     | ns   |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                | 20               |        |     | 24                |     | ns   |
|                   |                               | $V_{CC} = 6.0\text{ V}$                                | 17               |        |     | 20                |     | ns   |
|                   |                               | $\overline{MR}$ LOW; see Fig. 8                        |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 2.0\text{ V}$                                | 75               |        |     | 90                |     | ns   |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                | 15               |        |     | 18                |     | ns   |
|                   |                               | $V_{CC} = 6.0\text{ V}$                                | 13               |        |     | 15                |     | ns   |
| $t_{su}$          | set-up time                   | D, An to $\overline{LE}$ ; see Fig. 9 and Fig. 10      |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 2.0\text{ V}$                                | 75               |        |     | 90                |     | ns   |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                | 15               |        |     | 18                |     | ns   |
|                   |                               | $V_{CC} = 6.0\text{ V}$                                | 13               |        |     | 15                |     | ns   |
| $t_h$             | hold time                     | D to $\overline{LE}$ ; see Fig. 9 and Fig. 10          |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 2.0\text{ V}$                                | 5                |        |     | 5                 |     | ns   |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                | 5                |        |     | 5                 |     | ns   |
|                   |                               | $V_{CC} = 6.0\text{ V}$                                | 5                |        |     | 5                 |     | ns   |
|                   |                               | An to $\overline{LE}$ ; see Fig. 9 and Fig. 10         |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 2.0\text{ V}$                                | 5                |        |     | 5                 |     | ns   |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                | 5                |        |     | 5                 |     | ns   |
|                   |                               | $V_{CC} = 6.0\text{ V}$                                | 5                |        |     | 5                 |     | ns   |
| $C_{PD}$          | power dissipation capacitance | $f = 1\text{ MHz}$ ; $V_I = \text{GND to } V_{CC}$ [4] |                  | 26     |     |                   |     | pF   |
| <b>EM74HCT259</b> |                               |                                                        |                  |        |     |                   |     |      |
| $t_{pd}$          | propagation delay             | D to Qn; see Fig. 5 [2]                                |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                |                  |        | 17  |                   | 20  | ns   |
|                   |                               | An to Qn; see Fig. 6 [2]                               |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                |                  |        | 17  |                   | 20  | ns   |
|                   |                               | $\overline{LE}$ to Qn; see Fig. 7 [2]                  |                  |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                |                  |        | 17  |                   | 20  | ns   |

# EM74HC259; EM74HCT259

8-bit addressable latch

| Symbol    | Parameter                     | Conditions                                                               | -40 °C to +85 °C |     |     | -40 °C to +125 °C |     | Unit |
|-----------|-------------------------------|--------------------------------------------------------------------------|------------------|-----|-----|-------------------|-----|------|
|           |                               |                                                                          | Min              | Typ | Max | Min               | Max |      |
| $t_{PHL}$ | HIGH to LOW propagation delay | $\overline{MR}$ to $Q_n$ ; see Fig. 8                                    |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  |                  |     | 17  |                   | 20  | ns   |
| $t_t$     | transition time               | see Fig. 7 [3]                                                           |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  |                  |     | 7   |                   | 9   | ns   |
| $t_w$     | pulse width                   | $\overline{LE}$ HIGH or LOW; see Fig. 7                                  |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  | 20               |     |     | 24                |     | ns   |
|           |                               | $\overline{MR}$ LOW; see Fig. 8                                          |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  | 31               |     |     | 38                |     | ns   |
| $t_{su}$  | set-up time                   | D, An to $\overline{LE}$ ; see Fig. 9 and Fig. 10                        |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  | 15               |     |     | 18                |     | ns   |
| $t_h$     | hold time                     | D to $\overline{LE}$ ; see Fig. 9 and Fig. 10                            |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  | 5                |     |     | 5                 |     | ns   |
|           |                               | An to $\overline{LE}$ ; see Fig. 9 and Fig. 10                           |                  |     |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                                  | 5                |     |     | 5                 |     | ns   |
| $C_{PD}$  | power dissipation capacitance | $f = 1\text{ MHz}$ ;<br>$V_I = \text{GND to } V_{CC} - 1.5\text{ V}$ [4] |                  | 28  |     |                   |     | pF   |

[1] Typical values are measured at nominal supply voltage ( $V_{CC} = 3.3\text{ V}$  and  $V_{CC} = 5.0\text{ V}$ ).

[2]  $t_{su}$  is the same as  $t_{PLH}$  and  $t_{PHL}$ .

[3]  $t_t$  is the same as  $t_{THL}$  and  $t_{TLH}$ .

[4]  $C_{PD}$  is used to determine the dynamic power dissipation ( $P_D$  in  $\mu\text{W}$ ).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$  where:

$f_i$  = input frequency in MHz;

$f_o$  = output frequency in MHz;

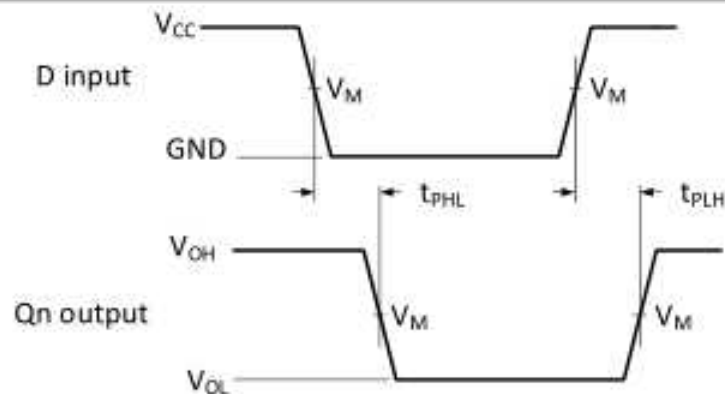
$C_L$  = output load capacitance in pF;

$V_{CC}$  = supply voltage in V;

$N$  = number of inputs switching;

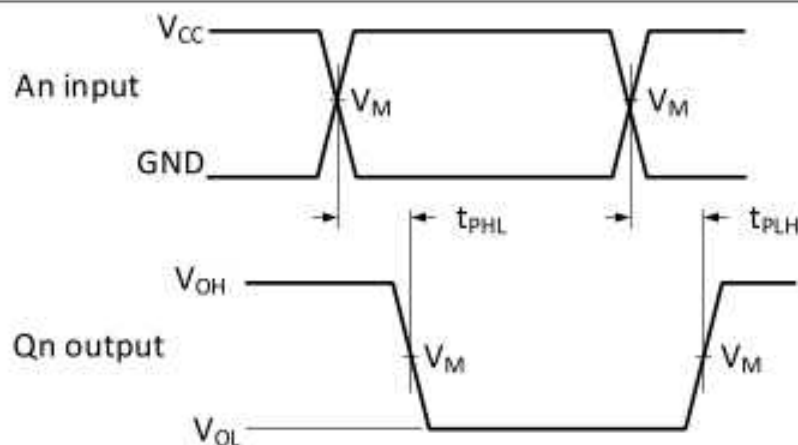
$\sum (C_L \times V_{CC}^2 \times f_o)$  = sum of outputs.

## 10.1. Waveforms and test circuit



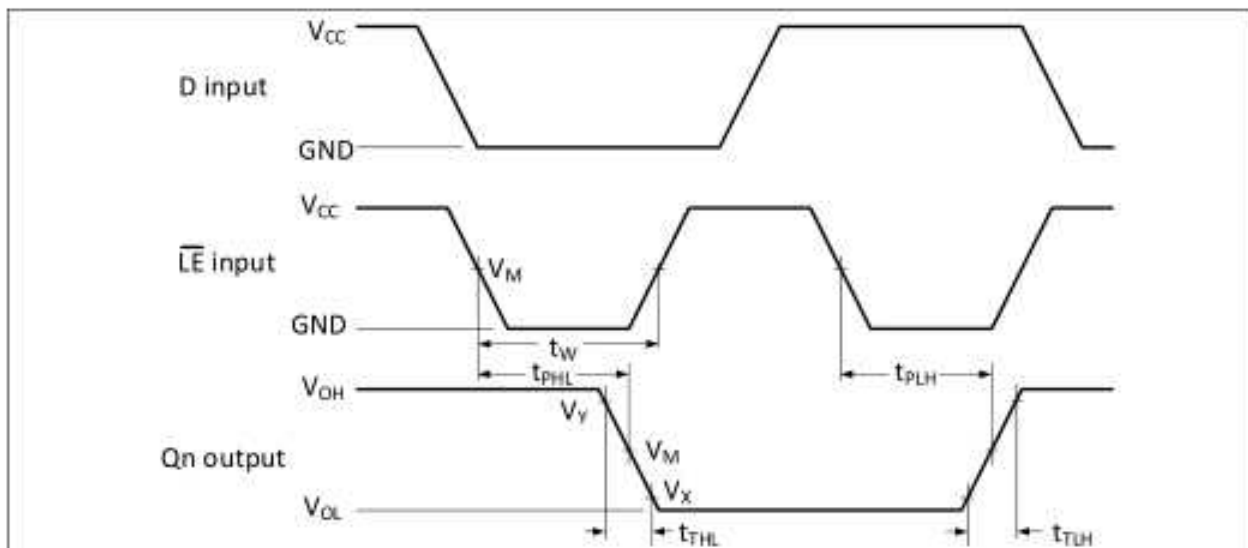
Measurement points are given in Table 8.  
 $V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 5. Data input to output propagation delays



Measurement points are given in Table 8.  
 $V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

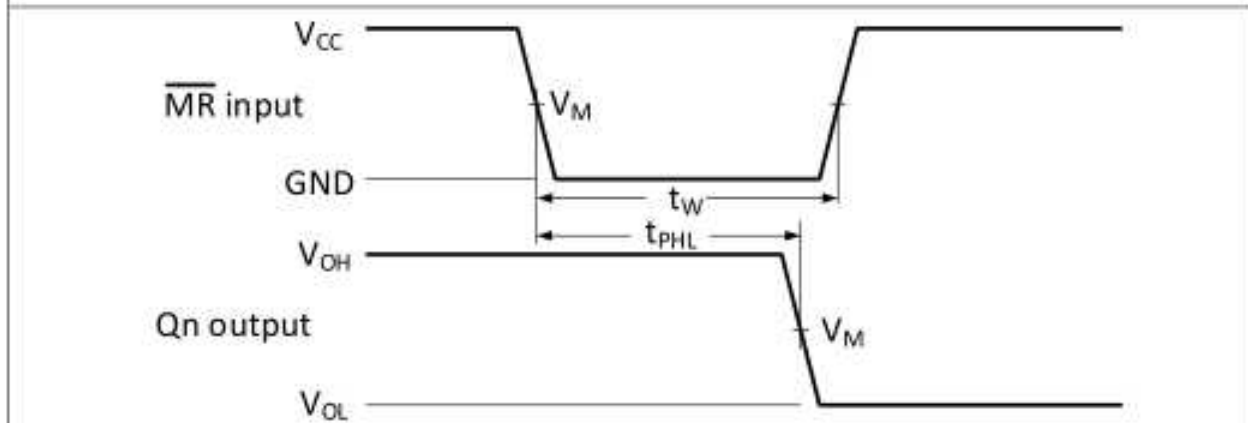
Fig. 6. Address input to output propagation delays



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical voltage output levels that occur with the output load.

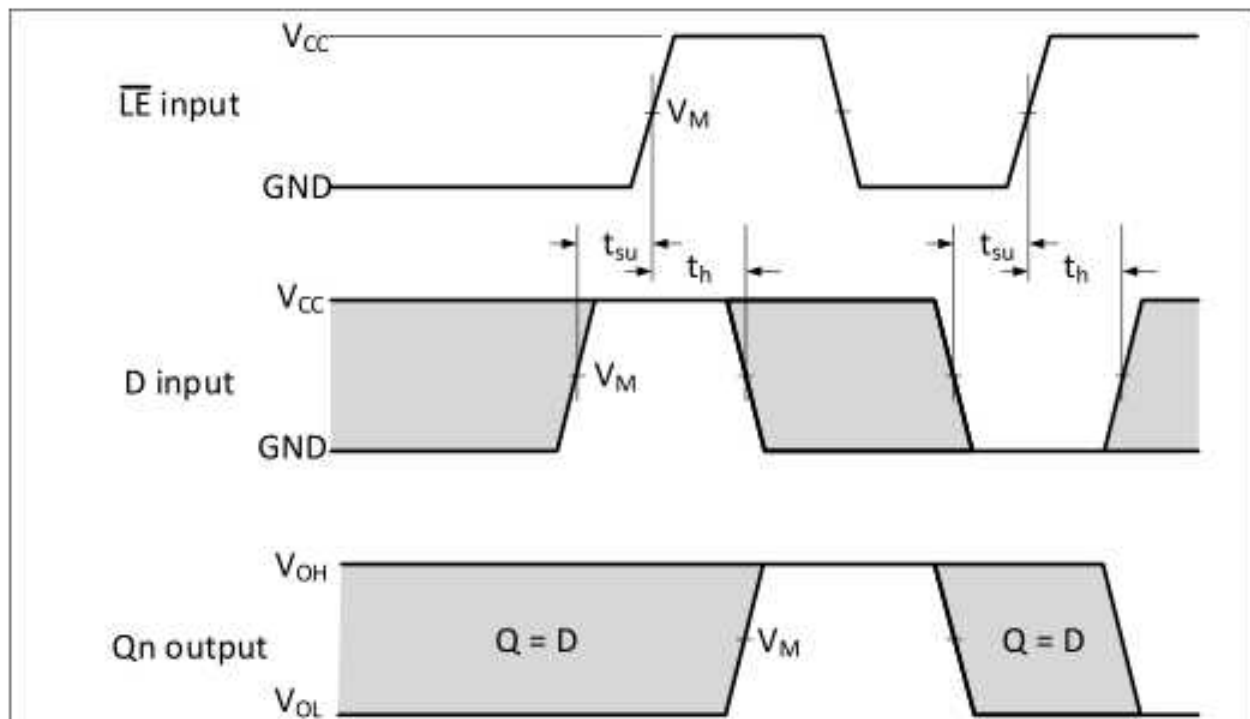
Fig. 7. Enable input to output propagation delays and pulse width



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical voltage output levels that occur with the output load.

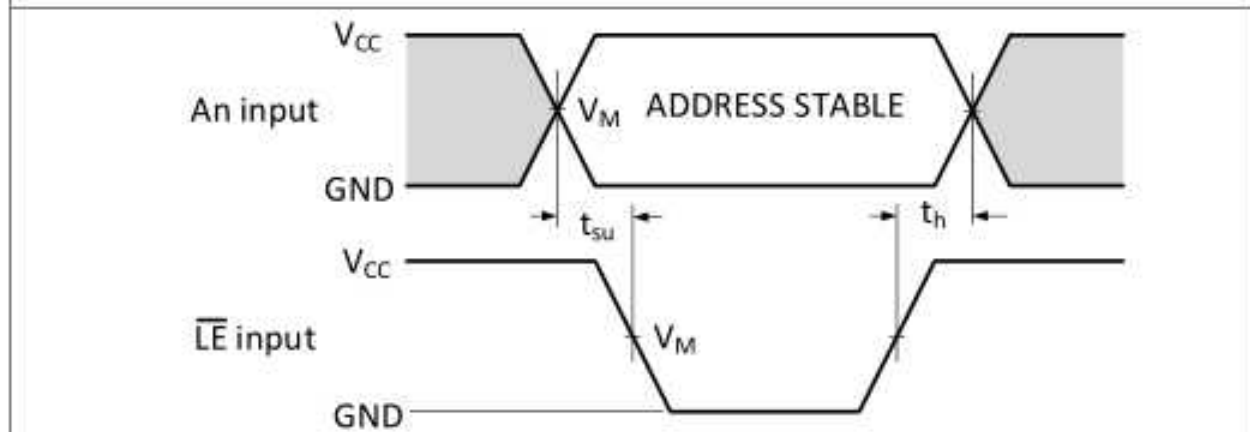
Fig. 8. Master reset input to output propagation delays



Measurement points are given in Table 8.

The shaded areas indicate when the input is permitted to change for predictable output performance.  $V_{OL}$  and  $V_{OH}$  are typical voltage output levels that occur with the output load.

Fig. 9. Data input to latch enable input set-up and hold times



Measurement points are given in Table 8.

The shaded areas indicate when the input is permitted to change for predictable output performance.  $V_{OL}$  and  $V_{OH}$  are typical voltage output levels that occur with the output load.

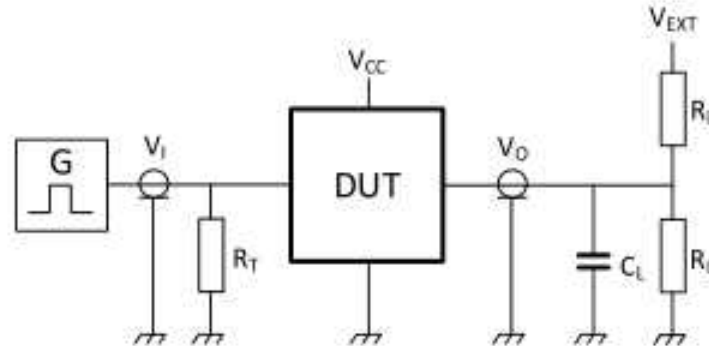
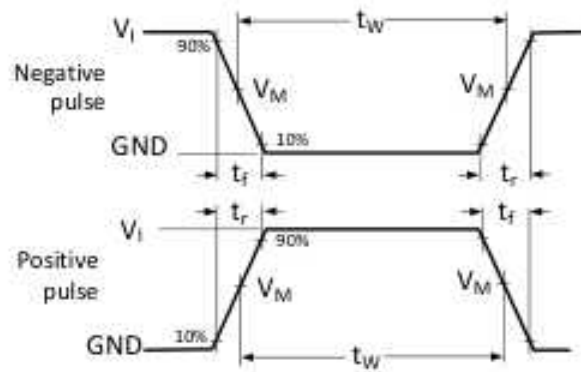
Fig. 10. Address input to latch enable input set-up and hold times

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Table 8. Measurement points

| Type       | Input       | Output      |             |             |
|------------|-------------|-------------|-------------|-------------|
|            | $V_M$       | $V_M$       | $V_X$       | $V_Y$       |
| EM74HC259  | $0.5V_{CC}$ | $0.5V_{CC}$ | $0.1V_{CC}$ | $0.9V_{CC}$ |
| EM74HCT259 | 1.3 V       | 1.3 V       | $0.1V_{CC}$ | $0.9V_{CC}$ |



Test data is given in Table 9.

Definitions for test circuit:

$R_L$  = Load resistance.

$C_L$  = Load capacitance including jig and probe capacitance.

$R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.

$V_{EXT}$  = External voltage for measuring switching times.

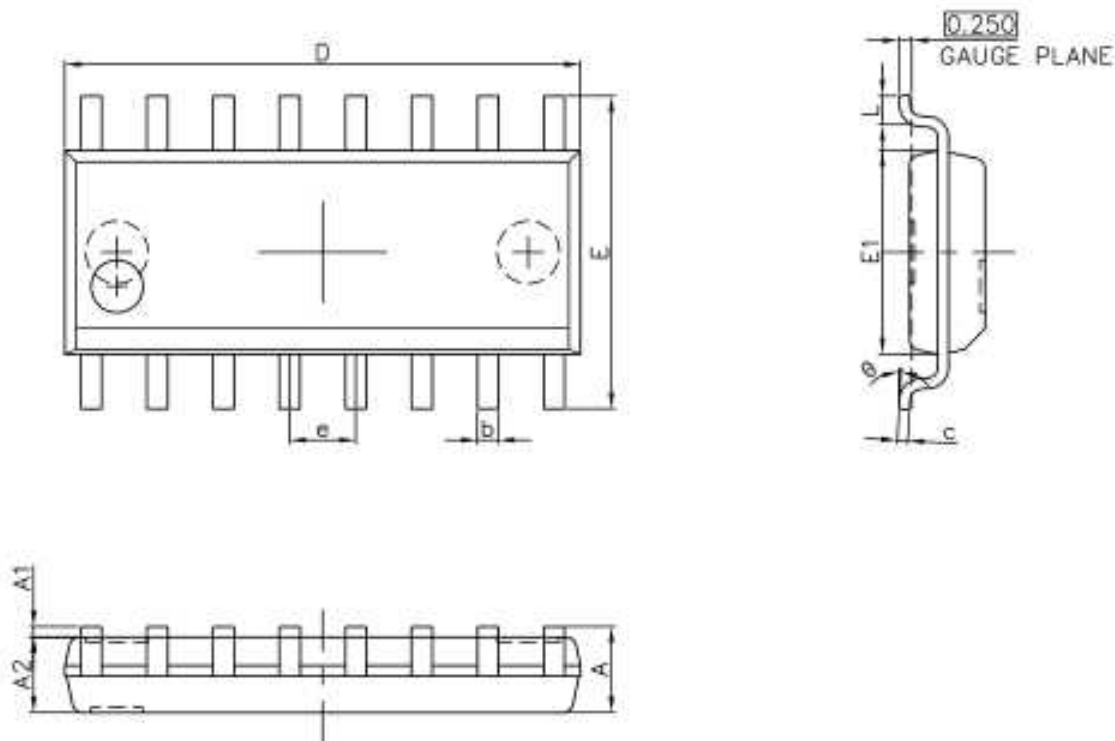
Fig. 11. Test circuit for measuring switching times

Table 9. Test data

| Type       | Input    |             | Load  |              | $V_{EXT}$          |
|------------|----------|-------------|-------|--------------|--------------------|
|            | $V_I$    | $t_r = t_f$ | $C_L$ | $R_L$        | $t_{PLH}, t_{PHL}$ |
| EM74HC259  | $V_{CC}$ | 2.0 ns      | 50 pF | 500 $\Omega$ | open               |
| EM74HCT259 | 3 V      | 2.0 ns      | 50 pF | 500 $\Omega$ | open               |

## 11. Package Outline

SOP-16L



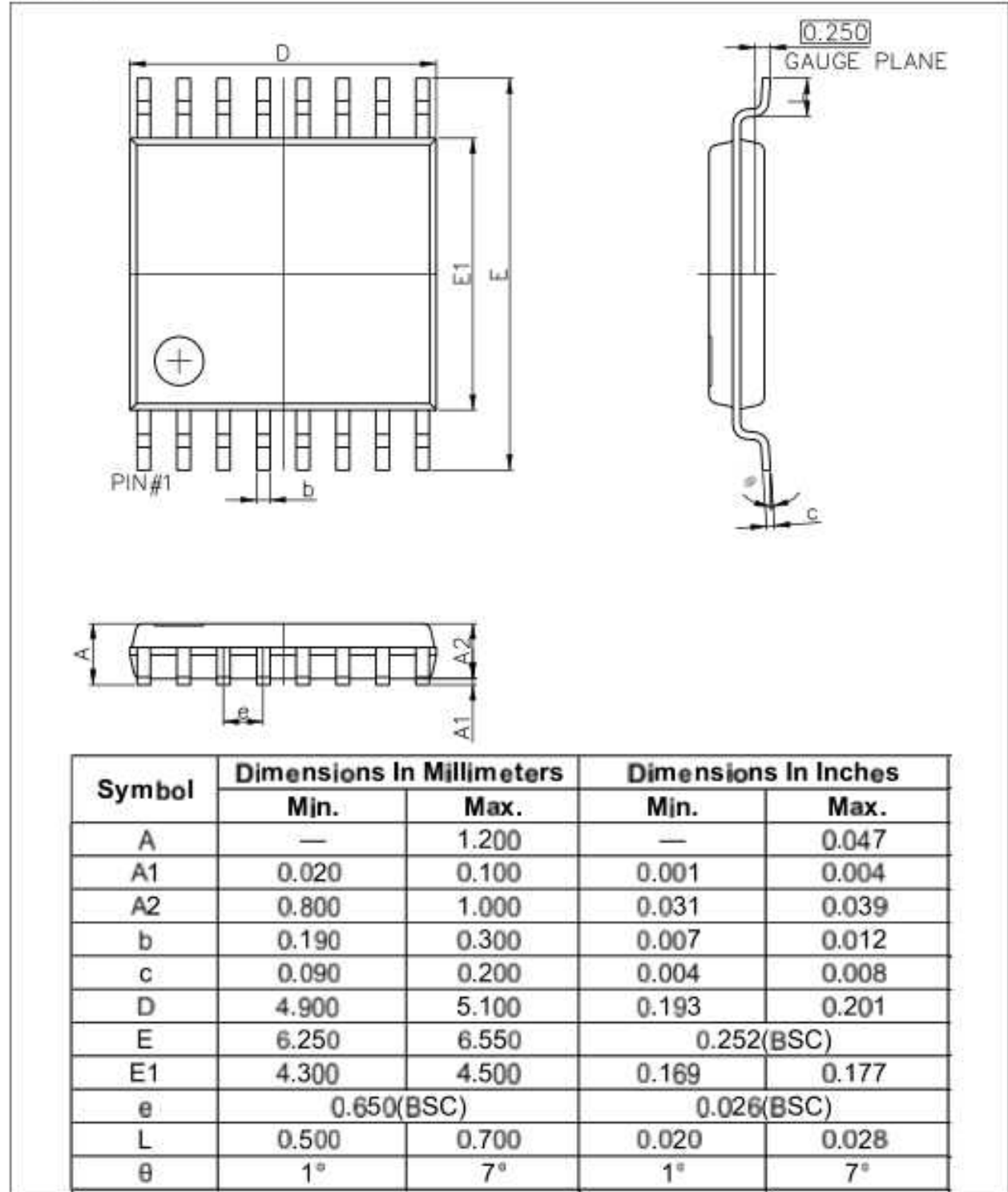
| Symbol | Dimensions In Millimeters |        | Dimensions In Inches |       |
|--------|---------------------------|--------|----------------------|-------|
|        | Min.                      | Max.   | Min.                 | Max.  |
| A      | —                         | 1.750  | —                    | 0.069 |
| A1     | 0.150                     | 0.250  | 0.006                | 0.010 |
| A2     | 1.400                     | 1.500  | 0.055                | 0.059 |
| b      | 0.330                     | 0.510  | 0.013                | 0.020 |
| c      | 0.170                     | 0.250  | 0.007                | 0.010 |
| D      | 9.800                     | 10.000 | 0.386                | 0.394 |
| e      | 1.270(BSC)                |        | 0.050(BSC)           |       |
| E      | 5.900                     | 6.100  | 0.232                | 0.240 |
| E1     | 3.800                     | 4.000  | 0.150                | 0.157 |
| L      | 0.400                     | 1.270  | 0.016                | 0.050 |
| θ      | 0°                        | 8°     | 0°                   | 8°    |



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## TSSOP-16L



## 12. Abbreviations

**Table 10. Abbreviations**

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| DUT     | Device Under Test                       |
| ESD     | ElectroStatic Discharge                 |
| HBM     | Human Body Model                        |
| CDM     | Charged Device Model                    |
| TTL     | Transistor-Transistor Logic             |

## 13. Revision History

**Table 11. Revision history**

| Document ID            | Release Date | Data sheet status | Change notice | Supersedes |
|------------------------|--------------|-------------------|---------------|------------|
| EM74HC_HCT259 Rev. 1.0 | Apr 28, 2025 | Product datasheet |               |            |